

## Unit 2: Abstract Algebra in Cryptography:

1. Prime Numbers.
2. Its factorization.
3. Arithmetic Modular.

### Properties of Modular Arithmetic

$$a \equiv km + b$$

$a$  is congruent to  $km + b$ .

$a$  is dividend  $m$  is the divisor,  $k$  is the quotient and  $b$  is the remainder.

It can also be written as

$$a \equiv b \pmod{m}$$

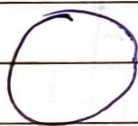
$$13 = 10 \times 1 + 3$$

$$13 \equiv 3 \pmod{10}$$

$$13 \equiv 3 \pmod{13} \quad \times$$

$$33 \equiv 3 \pmod{10} \quad \checkmark$$

$$43 \equiv 3 \pmod{10} \quad \checkmark$$



length of thread = 43

$$43 \equiv 4 \pmod{13}$$

circumference  
= 13

- Property 1:  $[(a \pmod{n}) + (b \pmod{n})] \pmod{n} = (a+b) \pmod{n}$
- Property 2:  $[(a \pmod{n}) - (b \pmod{n})] \pmod{n} = (a-b) \pmod{n}$
- Property 3:  $[(a \pmod{n}) \times (b \pmod{n})] \pmod{n} = (a \times b) \pmod{n}$

Example:  $A=15$ ,  $B=11$ ,  $n=8$

$$[(15 \pmod{8}) + (11 \pmod{8})] \pmod{8}$$

$$\Rightarrow (7+3) \pmod{8}$$

$$\Rightarrow 10 \pmod{8}$$

$$= 2$$

$$(15+11) \pmod{8} = 26 \pmod{8} = 2$$

$$[(15 \bmod 8) - (11 \bmod 8)] \bmod 8$$

$$[(7-3)] \bmod 8$$

$$= 4 \bmod 8$$

$$= 4.$$

$$[15-11] \bmod 8$$

$$= 4 \bmod 8$$

$$= 4.$$

• Commutative laws :

$$(a+b) \bmod n = (b+a) \bmod n$$

$$(a \times b) \bmod n = (b \times a) \bmod n$$

• Associative

$$((a+b)+c) \bmod n = (a+(b+c)) \bmod n$$

$$((a \cdot b) \cdot c) \bmod n = (a \cdot (b \cdot c)) \bmod n$$

• Distributive

$$(a(b+c)) \bmod n = (a \cdot b + a \cdot c) \bmod n.$$

$$= (a \cdot b) \bmod n + (a \cdot c) \bmod n.$$

• Identities

$$(0+a) \bmod n = (a+0) \bmod n = a \bmod n$$

$$(1 \cdot a) \bmod n = (a \cdot 1) \bmod n = a \bmod n$$

• Additive Inverse :

$\forall a$  there exist  $-a$  such that

$$a + (-a) \equiv 0 \bmod n \quad \text{where } a \in \mathbb{Z}_n$$

Q  $A=14, B=9, n=5$ , Solve using all Modulo arithmetic properties.

(C is 11.) where needed.

$$1. [(a \bmod n) + (b \bmod n)] \bmod n = (a+b) \bmod n$$

$$\begin{aligned} \text{LHS} &= [14 \bmod 5] + [9 \bmod 5] \bmod 5 \\ &= [4 + 4] \bmod 5 \\ &= 3 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= [14+9] \bmod 5 \\ &= [23] \bmod 5 \\ &= 3. \end{aligned}$$

$$\begin{aligned} 2. [a \bmod n] - [b \bmod n] \bmod n &= (a-b) \bmod n \\ [(14 \bmod 5)] - [9 \bmod 5] \bmod 5 &= (14-9) \bmod 5 \\ [4 - 4] \bmod 5 &= 5 \bmod 5 \\ 0 &= 0 \end{aligned}$$

$$\begin{aligned} 3. [(a \bmod n) \cdot (b \bmod n)] \bmod n &= (a \cdot b) \bmod n \\ [(4 \cdot 4) \bmod n] &= [14 \times 9] \bmod 5 \\ 1 &= 1. \end{aligned}$$

4. Commutative

$$\begin{aligned} (a+b) \bmod n &= (b+a) \bmod n \\ (14+9) \bmod 5 &= (9+14) \bmod 5 \\ (23) \bmod 5 &= (23) \bmod 5 \\ 3 &= 3 \end{aligned}$$

$$\text{LHS} = \text{RHS}$$

5. Associative

$$\begin{aligned} ((a+b)+c) \bmod n &= (a+(b+c)) \bmod n \\ ((14+9)+11) \bmod 5 &= (14+(9+11)) \bmod 5 \\ (23+11) \bmod 5 &= (14+20) \bmod 5 \\ 34 \bmod 5 &= 34 \bmod 5 \\ 4 &= 4 \end{aligned}$$

$$\text{LHS} = \text{RHS}$$

★ GCD (HCF)

$$\text{GCD}(12, 33)$$
$$= 3.$$

|   |    |    |    |
|---|----|----|----|
| 2 | 12 | 3  | 33 |
| 2 | 6  | 11 | 11 |
| 3 | 3  | 1  | 1  |

### \* Euclidean / Euclid Algorithm.

- Method 1 :

Euclidean Algorithm is used to find (GCD) for very large numbers.

Find GCD (12, 13) using Euclidean Algorithm.

$$A > B$$

|                 | Q | A  | B  | R    |
|-----------------|---|----|----|------|
| $A! > B$ (Swap) |   | 12 | 33 |      |
| S1:             | 2 | 33 | 12 | 9(A) |
| S2:             | 1 | 12 | 9  | 3    |
| S3:             | 3 | 9  | 3  | 0    |
|                 |   | 3  | 0  |      |

↑  $\text{gcd of } 12, 33$

- Find the  $\text{GCD}(750, 900)$

| Q    | A   | B   | R    |
|------|-----|-----|------|
| Swap | 750 | 900 |      |
| 1    | 900 | 750 | 1500 |
| 5    | 750 | 150 | 0    |
|      | 150 | 0   |      |

Find the GCD(252, 105)

| Q | A   | B   | R  |
|---|-----|-----|----|
| 2 | 252 | 105 | 42 |
| 2 | 105 | 42  | 21 |
| 2 | 42  | 21  | 0  |
|   | 21  | 0   |    |

Method

(2)

$a > b$

Euclid. GCD(a, b)

if  $b = 0$  then

return a

else

return Euclid. GCD(b,  $a \bmod b$ )

Method 2 (Algorithmic Approach)

Example: GCD(150, 12)

$$\begin{aligned}
 \text{GCD}(150, 12) &= \text{GCD}(12, 150 \bmod 12) \\
 &= \text{GCD}(12, 2) \\
 &= \text{GCD}(2, 12 \bmod 2) \\
 &= \text{GCD}(2, 0) \\
 &= 2
 \end{aligned}$$

Ex 2: GCD(83, 19)

$$\begin{aligned}
 \text{GCD}(83, 19) &= \text{GCD}(19, 83 \bmod 19) \\
 &= \text{GCD}(19, 7) \\
 &= \text{GCD}(7, 19 \bmod 7) \\
 &= \text{GCD}(7, 5) \\
 &= \text{GCD}(5, 7 \bmod 5)
 \end{aligned}$$

$$\begin{aligned}
 &= \text{GCD}(5, 2) \\
 &= \text{GCD}(2, 5 \bmod 2) \\
 &= \text{GCD}(2, 1) \\
 &= \text{GCD}(1, 2 \bmod 1) \\
 &= \text{GCD}(1, 0) \\
 &= \underline{1}
 \end{aligned}$$

If GCD of two numbers is 1, they are called relatively prime numbers or co-prime numbers.

### \* Relatively Prime Numbers (Co-Prime)

1. Scenario 1:  $A=11$ ,  $B=17$  (Both are prime)

$$\begin{aligned}
 \text{GCD}(17, 11) &= \text{GCD}(11, 6) = \text{GCD}(6, 5) = \text{GCD}(5, 1) = \text{GCD}(1, 0) \\
 &= \underline{1}
 \end{aligned}$$

2. Scenario 2:  $A=11$ ,  $B=21$  (One is prime)

$$\text{GCD}(21, 11) = \text{GCD}(11, 10) = \text{GCD}(10, 1) = \text{GCD}(1, 0) = \underline{1}$$

3. Scenario 3:  $A=12$ ,  $B=77$  (None is prime)

$$\text{GCD}(77, 12) = \text{GCD}(12, 5) = \text{GCD}(5, 2) = \text{GCD}(2, 1) = \text{GCD}(1, 0) = \underline{1}$$

Q1. Are 4 and 13 Co-prime.

$$\text{GCD}(13, 4) = \text{GCD}(4, 1) = \text{GCD}(1, 0) = \underline{1}$$

⇒ Yes, They are co-prime as  $\text{GCD}(13, 4) = 1$ .

Q2. Are (15, 21) relatively prime numbers

$$\text{GCD}(21, 15) = \text{GCD}(15, 6) = \text{GCD}(6, 3) = \text{GCD}(3, 0) = 3$$

No, They are not coprime, (relatively prime)  
as  $\text{GCD}(15, 21) \neq 1$

Q Find GCD of two no also find if they are co prime & find MI

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## \* Extended Euclidean's Algorithm.

- What is multiplicative inverse of 3 mod 5  
 $\Rightarrow$  A number that should be multiplied to 3  
 So that answer is 1

It is used to find multiplicative inverse (MI) using euclidean algorithm.

Q Find MI of 11 mod 13

| Q | A  | B  | R | T <sub>1</sub> | T <sub>2</sub> | T   |
|---|----|----|---|----------------|----------------|-----|
| 1 | 13 | 11 | 2 | 0              | 1              | -1  |
| 5 | 11 | 2  | 1 | 1              | -1             | 6   |
| 2 | 2  | 1  | 0 | -1             | 6              | -13 |
|   | 1  | 0  |   | 6              | -13            |     |

$\swarrow$  GCD
 $\swarrow$  MI

Initially value of  $T_1$  is 0 and  $T_2$  is 1

$$T = T_1 - T_2 \times Q$$

Q Find MI of 11 mod 26

| Q | A  | B  | R | T <sub>1</sub> | T <sub>2</sub> | T  |
|---|----|----|---|----------------|----------------|----|
| 2 | 26 | 11 | 4 | 0              | 1              | -2 |
| 2 | 11 | 4  | 3 | 1              | -2             | 5  |
| 1 | 4  | 3  | 1 | -2             | 5              | -7 |
| 3 | 3  | 1  | 0 | 5              | -7             | 26 |
|   | 1  | 0  |   | -7             | 26             |    |

$\swarrow$  GCD
 $\swarrow$  MI

MI must not be negative So add mod value

$-7 + 8 \times 26$  till it is +ve

$$\Rightarrow \boxed{MI = 19}$$

\* Euler's totient function & theorem.

It is denoted by  $\phi(n)$ . It is number of positive integers less than  $n$  that are relatively prime to  $n$ .

a Find  $\phi(5)$ .

$$n = 5.$$

Number less than 5 are: 1, 2, 3, 4.

$$\text{GCD}(1, 5) = 1$$

$$\text{GCD}(2, 5) = 1$$

$$\text{GCD}(3, 5) = 1$$

$$\text{GCD}(4, 5) = 1$$

$$\underline{\phi(5) = 4}$$

a Find  $\phi(11)$ .

Number less than 10: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

$$\text{GCD}(1, 11) = 1$$

$$(2, 11) = 1$$

$$(10, 11) = 1$$

$$\phi(11) = 10$$

Q  $\phi(8)$

Numbers less than 8 : 1, 2, 3, 4, 5, 6, 7

$\text{GCD}(1, 8) = 1$

$\text{GCD}(2, 8) = 2$

$\text{GCD}(3, 8) = 1$

$\text{GCD}(4, 8) = 4$

$\text{GCD}(5, 8) = 1$

$\text{GCD}(6, 8) = 2$

$\text{GCD}(7, 8) = 1$

$\phi(8) = 4$

| $\lambda$ | Criteria of 'n'            | Formula                                                                                  |
|-----------|----------------------------|------------------------------------------------------------------------------------------|
|           | n is prime.                | $\phi(n) = n - 1$                                                                        |
|           | $n = p \times q$           | $\phi(n) = (p-1) \times (q-1)$                                                           |
| $\phi(n)$ | p & q are prime            |                                                                                          |
|           | $n = a \times b$           |                                                                                          |
|           | either a or b is composite | $\phi(n) = n \times \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots$ |
|           | Both a & b are composite.  | where $p_1$ & $p_2$ are distinct                                                         |

1.  $\phi(5) = 5 - 1 = 4$  (as n is prime)

2.  $\phi(11) = 11 - 1 = 10$  (as n is prime)

3.  $\phi(35) : 35 = 5 \times 7$   
(both are prime)

$$\begin{array}{r|l} 5 & 35 \\ 7 & 7 \\ & 1 \end{array}$$

Thus,  $\phi(35) = (5-1) \times (7-1)$   
 $= 4 \times 6$   
 $= 24$

4.  $\phi(1000)$

$$1000 = 2^3 \times 5^3$$

$$\phi(n) = n \times \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right)$$

$$= 1000 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 1000 (0.5) (0.8)$$

$$= 400$$

$$\begin{array}{r|l} 2 & 1000 \\ 2 & 500 \\ 2 & 250 \\ 5 & 125 \\ 5 & 25 \\ 5 & 5 \\ 5 & 1 \end{array}$$

$$0.2$$

$$5 \sqrt{10}$$

$$0.1312$$

$$\begin{array}{r} 7 \sqrt{10} \\ 70 \\ \underline{30} \\ 30 \\ \underline{20} \\ 10 \\ \underline{10} \\ 0 \end{array}$$

$$0.8571312$$

$$\begin{array}{r} 7 \sqrt{60} \\ 56 \\ \underline{40} \\ 35 \\ \underline{50} \\ 29 \\ \underline{10} \end{array}$$

5.  $\phi(7000)$

$$7000 = 2^3 \times 5^3 \times 7$$

$$= 7000 (0.5) (0.8) (0.8571312)$$

$$= 2400$$

### \* Chinese Remainder Theorem

CRT is used to Solve a Set of different congruent equations with one variable but different moduli which are relatively prime as shown below.

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$x \equiv a_n \pmod{m_n}$$

CRT states that the above equation have a unique solution if the moduli are relatively prime.

$$x = (a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + \dots + a_n M_n M_n^{-1}) \pmod{M}$$

$$M = m_1 \times m_2 \times m_3 \dots$$

$$M_i = \frac{M}{m_i}$$

Solve the following equation using CRT

$$x \equiv 2 \pmod{3}$$

$$x \equiv 5 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

$$a_1 = 2, m_1 = 3$$

$$a_2 = 5, m_2 = 5$$

$$a_3 = 2, m_3 = 7$$

$$x = (a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + a_3 M_3 M_3^{-1}) \pmod{M}$$

\* To Find:  $M_1, M_2, M_3, M_1^{-1}, M_2^{-1}, M_3^{-1}, M$

$$M = m_1 \times m_2 \times m_3 = 3 \times 5 \times 7 = 3 \times 35 = 105$$

$$M_1 = \frac{M}{m_1} = 35$$

$$M_2 = \frac{105}{5} = 21$$

$$M_3 = \frac{105}{7} = 15$$

a)  $M_1 \times M_1^{-1} = 1 \pmod{m_1}$

$$35 \cdot M_1^{-1} = 1 \pmod{3}$$

$$M_1^{-1} = 2$$

b)  $M_2 \times M_2^{-1} = 1 \pmod{m_2}$

$$21 \times M_2^{-1} = 1 \pmod{5}$$

$$M_2^{-1} = 1$$

c)  $M_3 \times M_3^{-1} = 1 \pmod{m_3}$

$$15 \times M_3^{-1} = 1 \pmod{7}$$

$$M_3^{-1} = 1$$

$$\begin{aligned}
 x &= (2 \times 35 \times 2 + 5 \times 21 + 2 \times 15) \bmod M \\
 &= 275 \bmod M \\
 &= 275 \bmod 105 \\
 &= \underline{\underline{65}}
 \end{aligned}$$

$$\begin{aligned}
 4x &\equiv 5 \bmod 9 \quad \longrightarrow \text{Multiply by } 4^{-1} \text{ b.s.} \\
 2x &\equiv 6 \bmod 20 \quad \longrightarrow \text{Multiply by } 2^{-1} \text{ b.s.}
 \end{aligned}$$

$$4x \equiv 9$$

$$\begin{aligned}
 x &\equiv 4^{-1} \bmod 9 \times 5 \bmod 9 \\
 x &\equiv 7 \times 5 \bmod 9 = 35 \bmod 9 = 8 \bmod 9
 \end{aligned}$$

$$2x \equiv 6 \bmod 20$$

$$2x \equiv 2 \times 3 \bmod 20$$

$$x \equiv 3 \bmod 20$$

$$x \equiv 8 \bmod 9$$

$$x \equiv 3 \bmod 20$$

$$a_1 = 8$$

$$m_1 = 9$$

$$a_2 = 3$$

$$m_2 = 20$$

$$M = 20 \times 9 = 180$$

$$M_1 = \frac{180}{9} = 20$$

$$M_2 = \frac{180}{20} = 9$$

★ Fermat's Little Theorem.

If  $p$  is a prime and  $a$  is a positive number not divisible by  $p$  then

$$a^{p-1} \equiv 1 \pmod{p}$$

Example 1: Does Fermat's theorem hold true for  $p=5$  and  $a=2$ .

$$a^{p-1} \equiv 1 \pmod{p}$$

$$2^4 \equiv 1 \pmod{5}$$

$$16 \equiv 1 \pmod{5}$$

$$16 \bmod 5 = 1$$

$\Rightarrow$  Fermat's theorem holds true for  $p=5$  and  $a=2$

2. Prove Fermat's theorem holds true for  $p=13$  and  $a=11$ .

$a$  is positive

$p$  is prime

11 is not divisible by 13.

$$a^{p-1} \equiv 1 \pmod{p}$$

$$11^{13-1} \equiv 1 \pmod{13}$$

$$11^{12} \equiv 1 \pmod{13}$$

$$(-2)^{4 \times 3} \equiv 1 \pmod{13}$$

$$3^3 \equiv 1 \pmod{13}$$

$$27 \equiv 1 \pmod{13}$$

( $\because 11$  is 2 less than 13)

$$27 \bmod 13 = 1$$

$\Rightarrow$  Yes.

3.5 Prove format's theorem does not hold for  $p=6$ ,  $a=2$ . (Put in formula & prove).

\* Factorizing format's algorithm.

10 It is used to factorize  $n$  where  $n = x \cdot y$  works well when  $x$  and  $y$  are close

$$n = x^2 - y^2 \quad \text{--- (1)}$$

$$x^2 = n + y^2 \quad \text{--- (2)}$$

$$15 \quad x = \sqrt{n + y^2} \quad \text{--- (3)}$$

1. Factor  $n = 187$ .

$$20 \quad x = \sqrt{n + y^2}$$

$$= \sqrt{187 + 1^2} = \sqrt{188} \neq \text{Integer}$$

$$= \sqrt{187 + 2^2} = \sqrt{191} \neq \text{Integer}$$

$$= \sqrt{187 + 3^2} = \sqrt{196} = 14$$

$$25 \quad x = 14 \quad y = 3$$

$$n = x^2 - y^2$$

$$n = (x+y)(x-y)$$

$$n = (14+3)(14-3)$$

$$\boxed{n = (17)(11)}$$

$$30 \quad = 187$$

2. Factorize  $n = 3233$ .

$$x = \sqrt{n + y^2}$$

$$= \sqrt{3233 + 1^2}$$

$$= 56.89 \neq \text{Integer}$$

$$= \sqrt{3233 + 2^2}$$

$$= 56.89 \neq \text{Integer}$$

$$= \sqrt{3233 + 3^2}$$

$$= 56.93 \neq \text{Integer}$$

$$= \sqrt{3233 + 4^2}$$

$$= 57 \quad \checkmark$$

$$x = 57 \quad y = 4$$

$$n = (x^2 - y^2)$$

$$n = (x + y)(x - y)$$

$$n = (57 + 4)(57 - 4)$$

$$n = 61 \times 53$$

Thus,  $3233 = 61 \times 53$

3.  $n = 3009$

\* Discrete logarithm

$$5^1 \bmod 17 = 5$$

easy

$$5^? \bmod 17 = 5$$

hard

To find all these values which are uniformly distributed is easy, but to find  $x$  from the given equation we need discrete logarithm formula.

Formula is,

$$g^x \bmod p$$

Example 1: Solve  $\log_2 9 \text{ mod } 11$ .

$$\log_2 9 \text{ mod } 11$$

$$p = 11$$

$$q = 2$$

$$n = 9$$

$$x \equiv q^n \text{ mod } p$$

$$a \equiv 2^n \text{ mod } 11$$

$n$  can be  $[0 - 10]$

$$\text{at } n = 1 \quad a \equiv 2^1 \text{ mod } 11 = 2 \neq 9$$

$$\text{at } n = 2 \quad a \equiv 4 \text{ mod } 11 = 4 \neq 9$$

$$\text{at } n = 3 \quad a \equiv 8 \text{ mod } 11 = 8 \neq 9$$

$$\text{at } n = 4 \quad a \equiv 16 \text{ mod } 11 = 5 \neq 9$$

$$\text{at } n = 5 \quad a \equiv 32 \text{ mod } 11 = 10 \neq 9$$

$$\text{at } n = \boxed{6} \quad a \equiv 64 \text{ mod } 11 = 9 \quad \checkmark$$

$$a \equiv 2^6 \text{ mod } 11$$

$$a \equiv 64 \text{ mod } 11$$

$$a \equiv 9$$

2.

find value of  $x$  in  $2^x \text{ mod } 7 = 4$ .

$$\text{at } x = 1 \quad 4 \equiv 2^1 \text{ mod } 7 = 2 \neq 4$$

$$\text{at } x = 2 \quad 4 \equiv 4 \text{ mod } 7 = 4 \quad \checkmark$$

$$\boxed{x = 2}$$

{ Cyclic (Groups cyclic & finite fields)  
test for primality

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Date : \_\_\_\_\_

1. Cyclic Groups, rings, groups & finite fields
  2. test for primality
- Remains  
Topic

## \* 5 Test for Primality:

### 1. Fermat's Primality test:

$a^p - a \rightarrow p$  is Prime if this is a multiple of  $p$  for all  $1 \leq a \leq p$

a Is 5 prime?

Step 1:

at  $a = 1$

$$1^5 - 1 \rightarrow 0$$

0 is multiple of 5

at  $a = 2$

$$2^5 - 2$$

$= 30$  is multiple of 5

at  $a = 3$

$$3^5 - 3$$

$$= 240 \checkmark$$

at  $a = 4$

$$4^5 - 4$$

$$= 1020 \checkmark$$

$\Rightarrow 5$  is prime

### 2. Miller-Rabin Composite / Prime:

Algorithm:

S1: Find  $n-1 = 2^k \times m$

S2: Choose 'a' Such that  $1 < a < n-1$

S3: Compute  $b_0 = a^m \pmod{n}$ , ...,  $b_i = b_{i-1}^2 \pmod{n}$

+1  $\rightarrow$  Composite

-1  $\rightarrow$  probably prime

It checks whether a specific property which is known to hold for prime values or for composite values under testing.

Q Is 561 prime or not?  
 $n = 561$ .

S1:  $n-1 = 2^k \times m$

$$560 = 2^4 \times 35$$

S2:  $a = 2$ .

S3:  $b_0 = a^m \pmod{n}$

$$\begin{aligned} b_0 &= 2^{35} \pmod{561} \\ &= 263 \end{aligned}$$

$$\begin{aligned} b_1 &= b_0^2 \pmod{n} \\ &= (263)^2 \pmod{561} \\ &= 166 \end{aligned}$$

$$\begin{aligned} b_2 &= (166)^2 \pmod{561} \\ &= 67 \end{aligned}$$

$$\begin{aligned} b_3 &= (67)^2 \pmod{561} \\ &= 1 \rightarrow \text{Composite} \end{aligned}$$

56

$$\frac{560}{2^1} = 280 \checkmark$$

$$\frac{560}{2^2} = 140 \checkmark$$

$$\frac{560}{2^3} = 70 \checkmark$$

$$\frac{560}{2^4} = 35 \checkmark$$

$$\frac{560}{2^5} = 17.5$$

Q Is 53  
Prime

### \* Group:

A Group  $G$  is denoted by  $\{G, \cdot\}$  is a Set  $\cdot$  under Some operation  $\{\cdot\}$

If it Satisfies CAIN properties

1. Closure -  $(a \cdot b) \in G$  if  $a \in G$  &  $b \in G$ .
2. Associative -  $((a \cdot b) \cdot c) = (a \cdot (b \cdot c))$
3. ~~Inverse~~ Identity -  $a \cdot I = I \cdot a = a$ .
3. Inverse -  $a \cdot a^{-1} = a^{-1} \cdot a = I$

### \* Abelian Group:

A Group is said to be abelian if it is already a group and commutative property is also Satisfied.

1.  $(a \cdot b = b \cdot a)$

### \* Cyclic Group, Ring, Fields, Finite Field.